
CPSC 304 Project Cover Page

Milestone #: 1

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Group Number: 38

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Description

The Domain of the Application: Our application sits in the domain of urban public transit network management. It covers the operational and administrative side of running a city's transportation grid: the physical layout of the network (transit routes, individual stations, and schedules) and the financial side of commuter ticketing and fare validation. In short, it's the logic that governs how a transportation authority moves people around a city while keeping accurate records of network capacity, passenger usage, and rider payments.

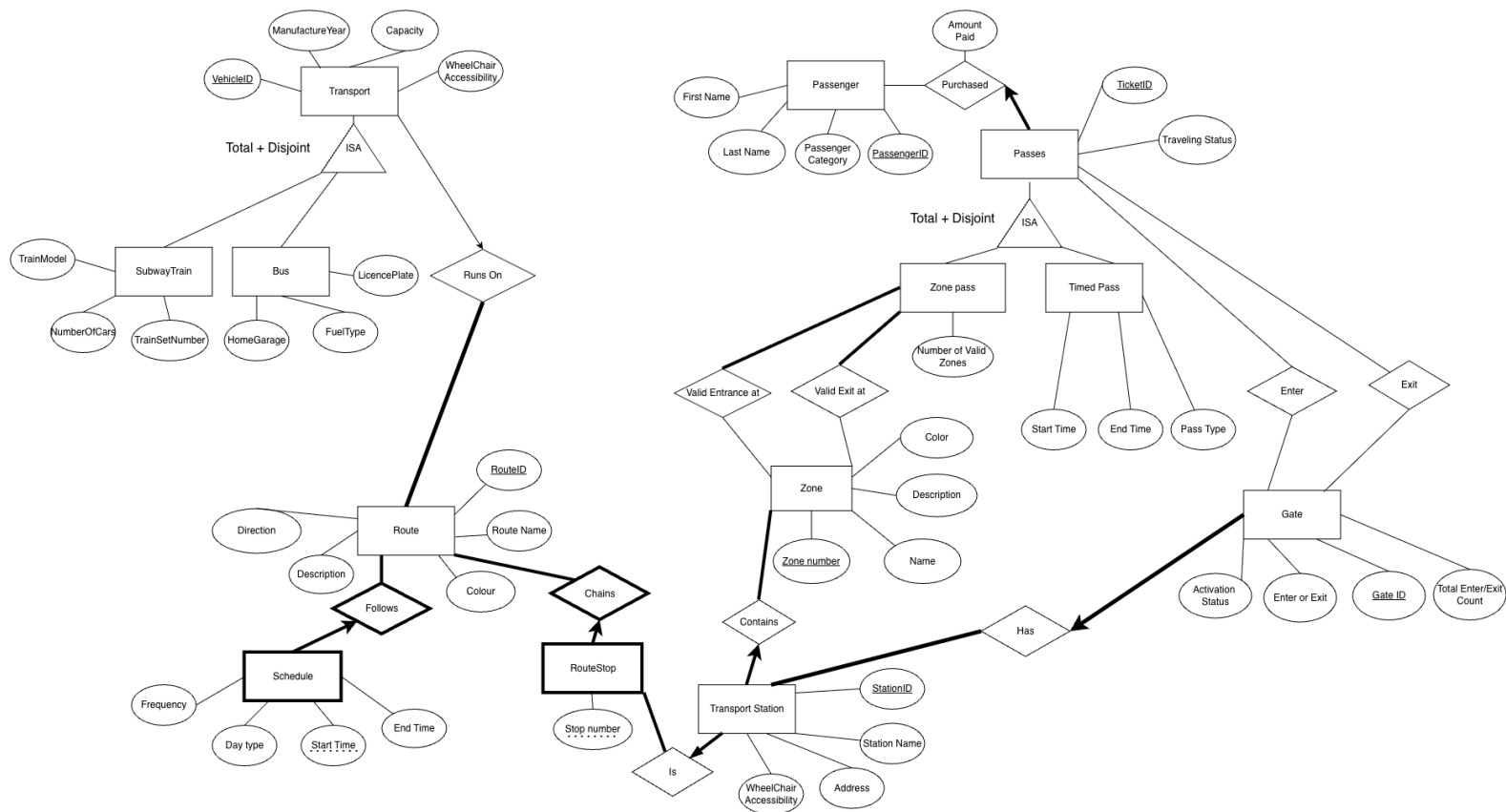
Aspects of the Domain Modeled by the Database: Our database focuses on the intersection of passenger ticketing and network routing. We deliberately left out peripheral operational data, like vehicle maintenance or human resources, and instead built around the core logic of how a commuter actually uses the grid.

On the network side, we model transit routes, the exact sequence of station stops, daily timetables, and the fare zones each station belongs to. On the passenger side, we model digital ticketing: commuter accounts, the pass types they can buy (timed unlimited passes, zone-specific cards, and so on), and the tap-in events that happen when they enter the system.

Think of TransLink here in Vancouver, or the TTC in Toronto, as a real-world parallel. When a commuter taps their card at a station gate, the application logs the entry, checks that their pass is active, confirms the pass covers that station's fare zone, and authorizes them to board. By narrowing the scope to these specific pieces, the database can handle complex queries around dynamic fare calculation, station traffic volume, and route sequencing over time.

Application Platform: Our application will run as a web-based platform built in JavaScript, matching the department's supported technology stack. On the backend, we're using Node.js to handle server logic, connecting to the department-provided Oracle database through standard drivers like oracledb to run our own hand-written SQL queries. We're not using any Object-Relational Mapping (ORM) libraries. The frontend will be built with standard HTML and JavaScript, without GUI generation tools. Every line of code and every SQL script will be written manually by our team, with no automated generation tools like GitHub Copilot. We'll track our progress through the version control repository CPSC 304 provided.

ER DIAGRAM:



Additional Comments:

Domain Differentiation and Approval: As reviewed and approved by the TA, our urban transit project differs fundamentally from the restricted airline domain. Airline databases center on individual point-to-point flights and single-use reservations. Our system models something different: continuous routes with ordered intermediate stops, recurring schedules, and mid-route transfers. Our ticketing structure also requires different logic. Instead of tracking specific flight segments, we handle time-based and geographical fare zones, covering things like daily passes, monthly passes, and stored-value transit cards.